

HW-2

Packages

Question 1

```
install.packages("aRtsy")
```

Install the `aRtsy` package

Question 2

```
library(aRtsy)
```

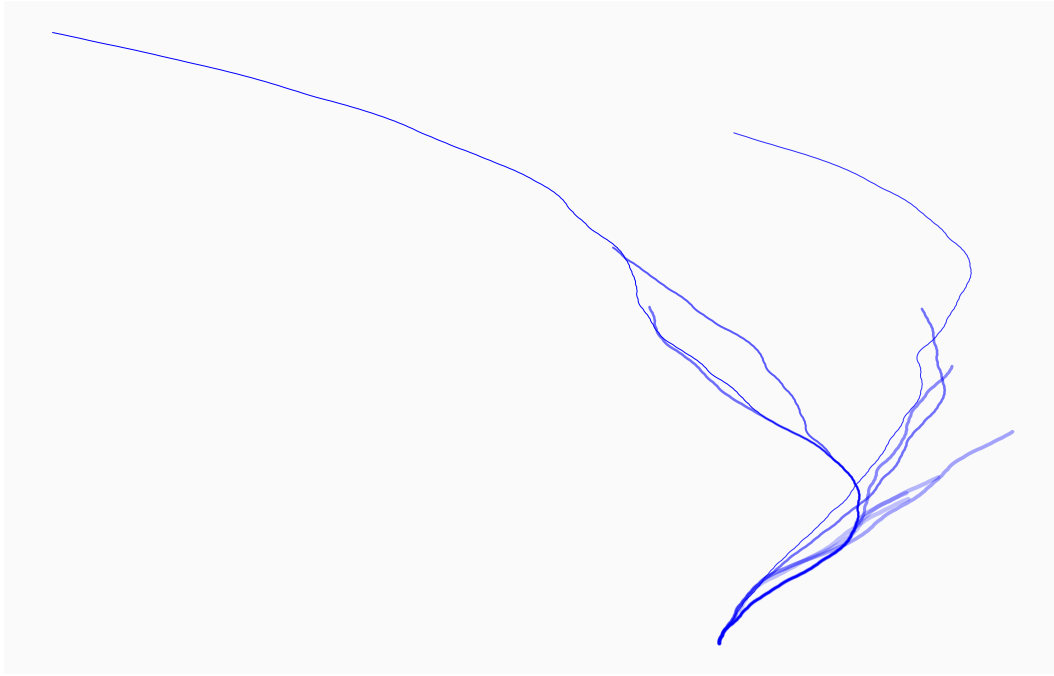
The Collatz conjecture starts with any positive integer n , if n is even, divide it by two; otherwise, multiply it by 3 and add 1. Repeats this process with the new n value. Eventually, n will reach to 1.

Let $f(n)$ be the next number obtained from n :
$$f(n) = \begin{cases} n/2, & \text{if } n/2 = 0, \\ 3n + 1, & \text{if } n/2 = 1. \end{cases}$$

Question 3

3.a

```
set.seed(7)
canvas_collatz(
  colors = c("blue"),
  n = 10
)
```



`canvas_collatz()` creates a plot of Collatz sequences using blue lines. `n=10` tells it to choose 10 random starting integers.

3.b

`canvas_collatz()` chooses 10 random starting integers, so it generates a different plot every time we run the code. This is troublesome in `.qmd` file because rendering again could change the plot, making the outcome hard to reproduce and compare. The `set.seed()` sets the same seed before generating random numbers gives the same outcome each time.